Alexander Bowring

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Summary

Senior Mathematical Consultant delivering advanced statistics, ML and optimisation solutions across major enterprises. Technical lead on £2M+ projects for clients including the National Energy System Operator, SSEN, and Coca-Cola, with deployed models demonstrated to save millions of pounds in operational costs. Combining advanced mathematics training (PhD Oxford, First-Class Mathematics Warwick) with commercial delivery expertise and cross-functional team leadership. My work has received national awards including the Operational Research Society Presidents Medal 2024.

Professional Experience

Senior Mathematical Consultant (previously Mathematical Consultant) — Oxford, UK

Smith Institute — *Nov 2021 – Present*

- **Technical Lead** across cross-functional teams (6–10 members), delivering **£2M+ project value** through advanced statistics, ML and optimisation solutions delivering multi-million pound operational savings for large-scale enterprise clients. Regularly engage with technical, commercial, and client stakeholders to translate complex methods into strategic business outcomes. Key projects include:

Dynamic Reserve Setting (DRS) – with the National Energy System Operator (NESO)

- Technical Lead for multi-year programme delivering end-to-end design and deployment of explainable gradient-boosted ML models to forecast national energy reserve requirements.
- Led the technical team in all aspects of data ingestion, model training and validation, managing hundreds of config-driven experiments using AWS and Azure DevOps for CI/CD and code review, and developed a Power BI dashboard enabling control room engineers to interact with model outputs.
- **Impact:** DRS models reduced reserve procurement requirements by on average 300MW every 30 minutes, saving NESO £1.7 million in reserve costs each month. Deployed in national control room. Awarded the Operational Research Society President's Medal 2024 for best application of OR.
- **Stack**: Python (pandas, LightGBM, Optuna, numpy, SHAP) | SQL | AWS (EC2, S3) | Power BI | Azure DevOps | bash.

Vulnerability Future Energy Scenarios (VFES) – with SSEN

- Principal model developer for project delivering explainable gradient-boosted ML models to identify common drivers of vulnerability in communities served by SSEN.
- Developed and trained gradient boosted models, ran experiments on cloud infrastructure, used ML SHAP explainability and advanced clustering to identify common sets of features driving vulnerability across communities.
- **Impact:** Results have shaped SSEN's network investment strategy for vulnerable communities and low-carbon transition planning. Awarded the Utility Week Unlocking Data Award 2024 and DataIQ Award 2024 for the Best Use of AI for the Public Good.
- **Stack:** Python (pandas, LightGBM, Optuna, sklearn, SHAP, matplotlib) | AWS (CodeCommit, EC2) | SQL.

Dispatch Decision Intelligence – with NESO

- Technical Lead for project delivering performance, functionality and analytics upgrades to optimisers used in the national control room for procuring national electricity generation.
- Designed explainable optimisation module with interactive interface and led technical team in its implementation, enabling control room engineers to interrogate and understand the rationale behind the optimiser's recommended generator movements in real-time.
- Designed and oversaw development of head and footroom module quantifying generation bounds with associated costs, supporting control room engineers to make decisions about expanding available capacity to mitigate against generation or demand forecast uncertainty.
- **Impact:** Improved optimiser run-times and enhanced decision transparency, key components are now being trialled across multiple national control room optimisers.
- **Stack:** Python (pandas, numpy) | Gurobi | AWS (S3, EC2) | Azure DevOps.

Weather Driven Sales Forecasting – with Coca-Cola European Partners (CCEP)

- Technical Lead for multi-year project developing predictive models to provide insights on weather-driven demand patterns across multiple drink categories and European markets for CCEP.
- Led technical overhaul of data ingestion pipeline following third-party data format changes, ensuring continued delivery of critical weather-sales insights.
- Parallelised model execution pipelines, reducing weekly deliverable processing time by 18%.
- **Impact:** Weekly reports actively used to guide CCEP's sales and distribution strategy across European markets.
- **Stack:** R (dplyr, ggplot2, tidyr, readr, officer) | Microsoft Excel and Powerpoint (advanced features) | Azure DevOps.

Research Fellow in Neuroimaging Statistics — Oxford, UK

Nuffield Department of Population Health, University of Oxford — *Nov 2019 - Nov 2021*

- Led research on fMRI analysis variability, implementing parallel analysis on 59 unique neuroimaging pipelines using Python, Matlab, and Unix on a high-performance computing cluster. Published research in Nature and Human Brain Mapping.
- Organised a weekly reading group featuring international experts in neuroimaging and taught statistical methods and Python programming to PhD students on the university's CDT programme.

Pre-PhD Research Roles — Coventry, UK

University of Warwick — *Jun 2015 - Oct 2016*

- **Research Assistant in Neuroimaging Statistics, Warwick Manufacturing Group**: Developed data-sharing standards for neuroimaging, performed fMRI analyses, and mentored a PhD student. Published research in Nature: Scientific Data and Frontiers in Neuroinformatics.
- **Undergraduate Research Intern**: Developed neuroimaging analysis pipelines to test fMRI visualisation software.

Education

Doctor of Philosophy in Population Health — Oxford, UK

Nuffield Department of Population Health, University of Oxford — *Oct 2016 - Nov 2019*

Thesis: On the Reproducibility and Interpretability of Group-Level Task-fMRI Results

- Developed novel Confidence Sets method for group-level inference on task-fMRI, advancing statistical theory in linear modelling and statistical bootstrapping. Demonstrated improved performance over traditional statistical methods on large-sample datasets and published research in NeuroImage.
- Assessed reproducibility of task-fMRI results by designing and implementing neuroimaging pipelines across multiple software packages. Developed Jupyter Notebooks to enable other researchers to reproduce the analyses and published research in Human Brain Mapping.
- Gave two oral presentations at the international Organisation for Human Brain Mapping (OHBM) conference (<5% of abstract submissions chosen for oral presentation). Recognised with £2000 Merit Abstract Award.
- Analysed big neuroimaging datasets including the UK Biobank and Human Connectome Project datasets.

Bachelor of Science in Mathematics — University of Warwick, UK

University of Warwick — *Sep 2012 - Jul 2015*

Graduated with **First Class Honours**, achieving >80% in modules including Algebra II, Analysis III, Galois Theory, Programming for Scientists, and Mathematics by Computer.

Skills

Machine Learning & Statistics: Gradient boosting (LightGBM, XGBoost), SHAP explainability, clustering, Optuna hyperparameter optimisation, reproducibility methods, statistical modelling/testing, experimental design
Optimisation: Gurobi, linear and mixed-integer programming, explainable optimisation
Programming & Data Analysis: Python (pandas, numpy, sklearn, Jupyter), R (tidyverse, ggplot2, dplyr), SQL, bash
Cloud & DevOps: AWS (EC2, S3, CodeCommit), MLOps, Azure DevOps, CI/CD pipelines, Git/Github
Data Visualisation & BI: Power BI, ggplot2, matplotlib, interactive dashboards, report automation

Publications

I have published academic research in several high-impact journals including *Nature*, *NeuroImage*, and *Human Brain Mapping*. Full list available at Google Scholar.